

Abhishek Lohar

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PROFILE

Highly motivated Computer Science and Engineering student at VIT-AP with a strong focus on Artificial Intelligence, Deep Learning, and backend development. Experienced in building AI-powered applications, RAG pipelines, full-stack systems, and scalable software solutions. Passionate about open-source development and solving real-world problems through technology.

EDUCATION

Vellore Institute of Technology-AP

B.Tech in Computer Science and Engineering — Current CGPA: **8.33 / 10.0**

Amaravati, AP

2023 – 2027

Vidya Bharati Chinmaya Vidyalaya (CBSE)

Jamshedpur, JH

Higher Secondary Education (Class XII) — Percentage: **79.8%**

2022

Vidya Bharati Chinmaya Vidyalaya (CBSE)

Jamshedpur, JH

Secondary Education (Class X) — Percentage: **85.8%**

2020

TECHNICAL SKILLS

Languages: Java, Python, SQL, C++, JavaScript, HTML/CSS

AI/ML Stack: LLMs, RAG, Embeddings, LangChain, ChromaDB, Ollama, OpenCV, Data Science

Backend & Tools: FastAPI, React, PostgreSQL, Redis, Kafka, Raspberry Pi, Git, pytest, PyPI, GitHub Actions, Open-Source

Soft Skills: Leadership, Creative Thinking, Problem Solving

PROJECTS

local-persona-memory — Published PyPI Package | *Python, Ollama, ChromaDB, LangChain, RAG*

- Published production-grade open-source Python package on PyPI (pip install local-persona-memory), giving local LLMs permanent long-term memory with zero cloud dependency.
- Built full RAG pipeline with ChromaDB vector storage, PDF ingestion, semantic search, conversation history, export/import, and callback hooks. Shipped with 160 passing tests and GitHub Actions CI/CD.

Fact Knowledge Layer — Document Intelligence System | *Python, FastAPI, OpenAI, ChromaDB, SQLite, React*

- Built enterprise document intelligence pipeline extracting atomic, auditable facts from financial PDFs with value, unit, time, scope, source quote, and page-level provenance using GPT-4o mini.
- Engineered two-stage reconciliation engine using ChromaDB semantic retrieval to prune irrelevant fact pairs before GPT-4o reasoning, reducing potential $O(N^2)$ comparisons by ~98%.
- Implemented FastAPI + React web application with asynchronous background processing, multi-dimensional fact filtering, evidence drill-down, and evaluation dashboard.

URL Shortener & Analytics Platform | *Python, FastAPI, PostgreSQL, Redis, Kafka*

- Built microservices-based URL shortener with separate Shorten, Redirect, and Analytics services behind an API gateway to practice service decomposition.
- Used PostgreSQL for persistent URL mappings and Redis caching in front of Redirect service; Kafka streams click events asynchronously to Analytics.

CERTIFICATIONS & ACHIEVEMENTS

- Oracle Cloud Infrastructure (OCI) Certified AI Foundations Associate
- Finalist — Engineering Clinics 2025
- High-Performance Coding (HPC) Certification by IAMNEO
- Infosys Springboard — Introduction to Data Science (May 2026)
- Hashgraph Developer Course — Hack2Skill GDG Hackathons at VIT-AP

LANGUAGES

Hindi (Native), English (Professional)